COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 1 : Session – 8 : SLO – 1 | Binary arithmetic and BCD operations
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I. Match the Following

• 1. Sign Magnitude representation — b) Separate bit for sign, remaining for magnitude
• 2. 1's Complement — e) Invert all bits of the subtrahend
• 3. 2's Complement — c) Invert all bits and add 1 to LSB
• 4. Binary Multiplication Rule — d) 1 × 1 = 1 (partial-product/shift-add rule)
• 5. BCD Addition — a) Add 0110 (6) for correction if sum > 9

II. Fill in the Blanks

• 1. sign 2. inverting (complementing) 3. 1 4. 1 5. 0110

III. Activity / Describe and Perform

• 1. Sign Magnitude — Addition: same signs → add magnitudes, keep sign (+3=0011, +2=0010 → 0101=+5). Subtraction (+5)-(+3): compare magnitudes, subtract smaller from larger (101-011=010), keep sign of larger → 0010 = +2.
• 2. 1's Complement — Addition: add all bits incl. sign, add end-around carry. Subtraction 7-2: 0111 + 1's of 0010 (1101) = 1 0100 → EAC → 0100+1 = 0101 = +5.
• 3. 2's Complement — Subtraction 7-2: 0111 + 2's of 0010 (1110) = 1 0101 → discard carry → 0101 = +5. No carry → negative result, take 2's complement for magnitude.
• 4. Multiplication 1011 × 101 (11×5): partial products by shift-and-add → 0110111 (55). Division 1010 ÷ 10 (10÷2): repeated shift-and-subtract (long division) → quotient 0101 (5), remainder 0.
• 5. BCD: each decimal digit → 4-bit code. Example 8+5: 1000+0101 = 1101 (>1001) → +0110 → 1 0011 → 0001 0011 = 13.